

Lucy McManamon

CS 4420

2 Oct. 2026

Assignment 05

Compile and Run

```
gcc -Wall -Werror assignment05.c -o assignment05
```

```
./assignment05 input [FCFS|RR|SJF] [time quantum]
```

- e.g.:

- ./assignment05 input FCFS
- ./assignment05 input SJF
- ./assignment05 input RR 5

Change line 32 to 1 for debug info

```
31 // debug flag
32 int d = 0;
```

Note: the program will not behave well if the process ids are not listed like 0, 1, ..., n-1. Please test them in this format.

Test Cases

All are also uploaded to GitHub under “test” folder.

1.

1	4
2	0 0 20
3	1 1 2
4	2 2 4
5	3 3 1

2.

1	4
2	0 0 8
3	1 0 2
4	2 0 5
5	3 0 3

3.

1	5
2	0 0 15
3	1 2 2
4	2 3 3
5	3 5 1
6	4 6 2

```
mcmam@pierre Assignment-05 % ./assignment05 test/test1.txt FCFS
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	20	20	0
1	1	20	22	2	19
2	2	22	26	4	20
3	3	26	27	1	23

Average Waiting Time = 15.50

```
mcmam@pierre Assignment-05 % ./assignment05 test/test1.txt SJF
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	20	20	0
3	3	20	21	1	17
1	1	21	23	2	20
2	2	23	27	4	21

Average Waiting Time = 14.50

```
mcmam@pierre Assignment-05 % ./assignment05 test/test1.txt RR 2
```

PID	Start Time	End Time	Running Time
0	0	2	2
1	2	4	2
2	4	6	2
0	6	8	2
3	8	9	1
2	9	11	2
0	11	13	2
0	13	15	2
0	15	17	2
0	17	19	2
0	19	21	2
0	21	23	2
0	23	25	2
0	25	27	2

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	27	20	7
1	1	2	4	2	1
2	2	4	11	4	5
3	3	8	9	1	5

Average Waiting Time = 4.50

```
● mcman@pierre Assignment-05 % ./assignment05 test/test2.txt FCFS
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	8	8	0
1	0	8	10	2	8
2	0	10	15	5	10
3	0	15	18	3	15

Average Waiting Time = 8.25

```
● mcman@pierre Assignment-05 % ./assignment05 test/test2.txt SJF
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
1	0	0	2	2	0
3	0	2	5	3	2
2	0	5	10	5	5
0	0	10	18	8	10

Average Waiting Time = 4.25

```
● mcman@pierre Assignment-05 % ./assignment05 test/test2.txt RR 3
```

PID	Start Time	End Time	Running Time
0	0	3	3
1	3	5	2
2	5	8	3
3	8	11	3
0	11	14	3
2	14	16	2
0	16	18	2

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	18	8	10
1	0	3	5	2	3
2	0	5	16	5	11
3	0	8	11	3	8

Average Waiting Time = 8.00

```
mcman@pierre Assignment-05 % ./assignment05 test/test3.txt FCFS
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	15	15	0
1	2	15	17	2	13
2	3	17	20	3	14
3	5	20	21	1	15
4	6	21	23	2	15

Average Waiting Time = 11.40

```
mcman@pierre Assignment-05 % ./assignment05 test/test3.txt SJF
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	15	15	0
3	5	15	16	1	10
1	2	16	18	2	14
4	6	18	20	2	12
2	3	20	23	3	17

Average Waiting Time = 10.60

```
mcman@pierre Assignment-05 % ./assignment05 test/test3.txt RR 4
```

PID	Start Time	End Time	Running Time
0	0	4	4
1	4	6	2
2	6	9	3
0	9	13	4
3	13	14	1
4	14	16	2
0	16	20	4
0	20	23	3

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	23	15	8
1	2	4	6	2	2
2	3	6	9	3	3
3	5	13	14	1	8
4	6	14	16	2	8

Average Waiting Time = 5.80